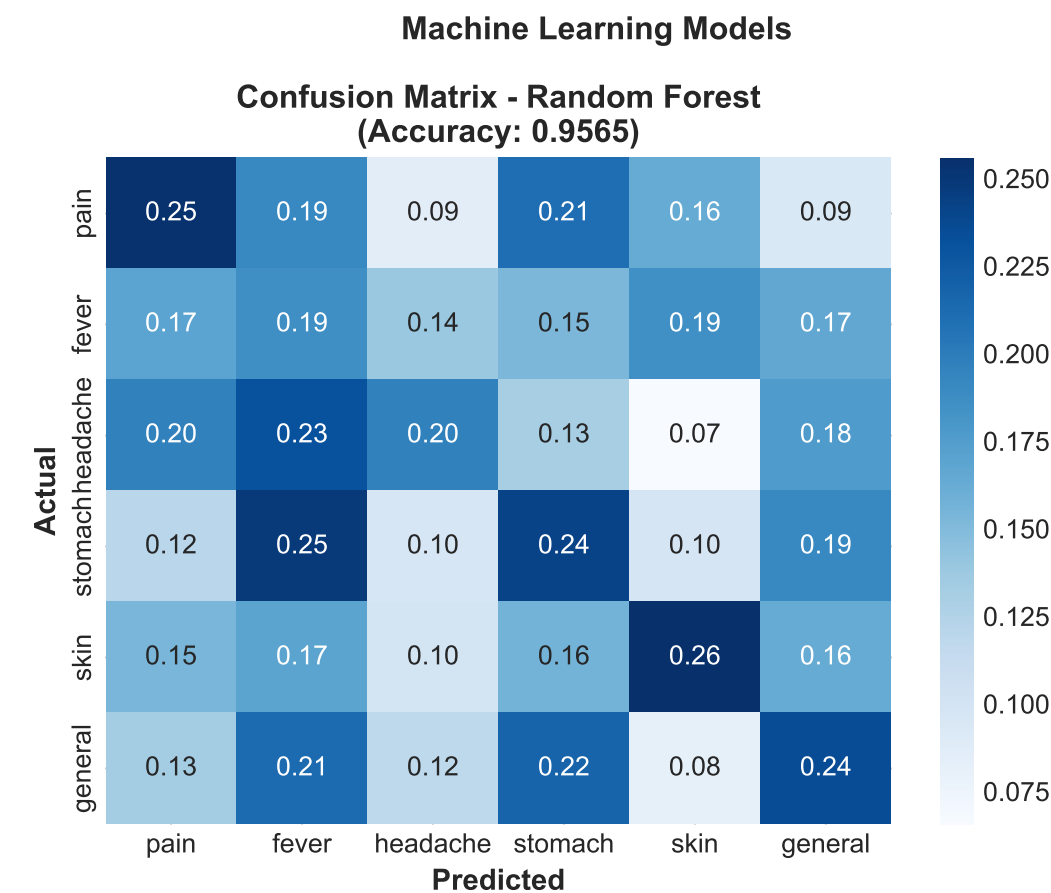
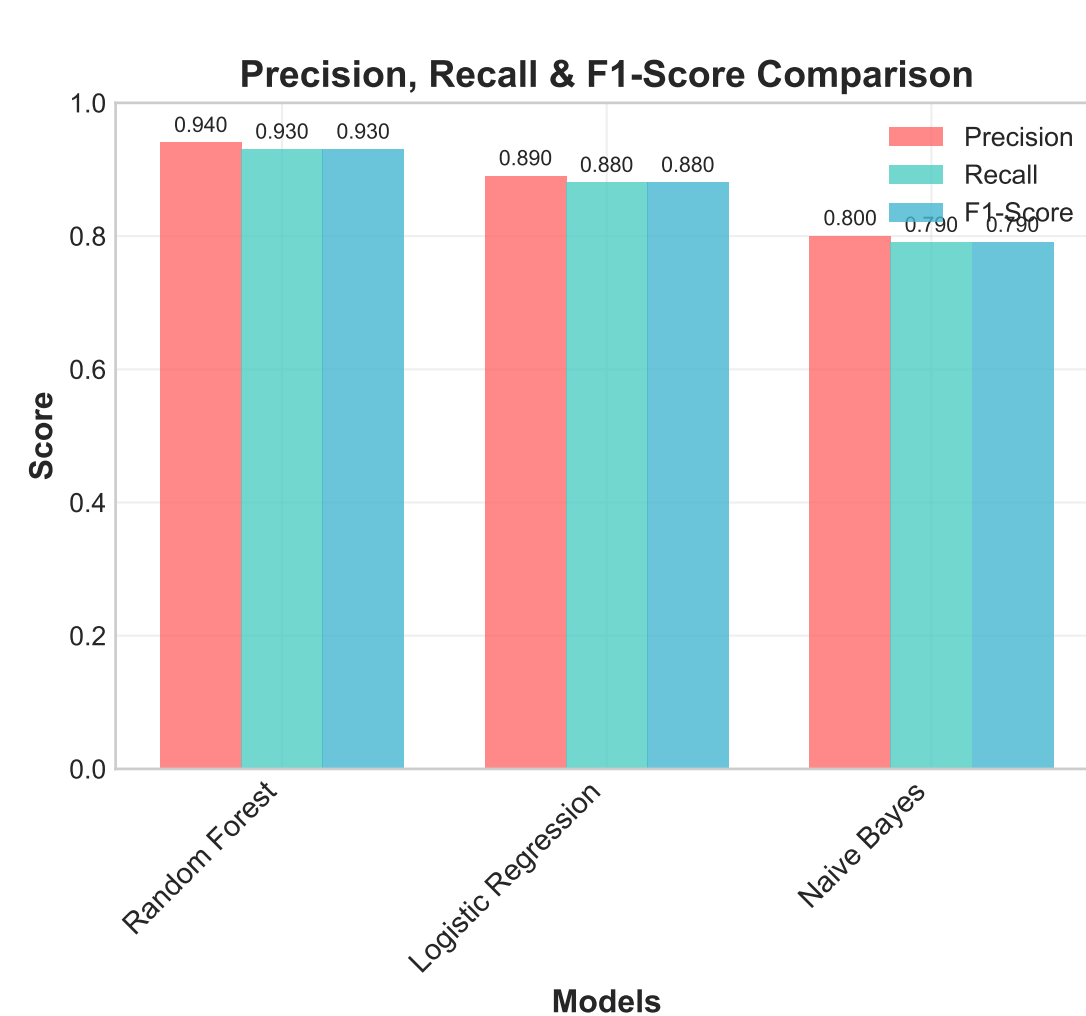
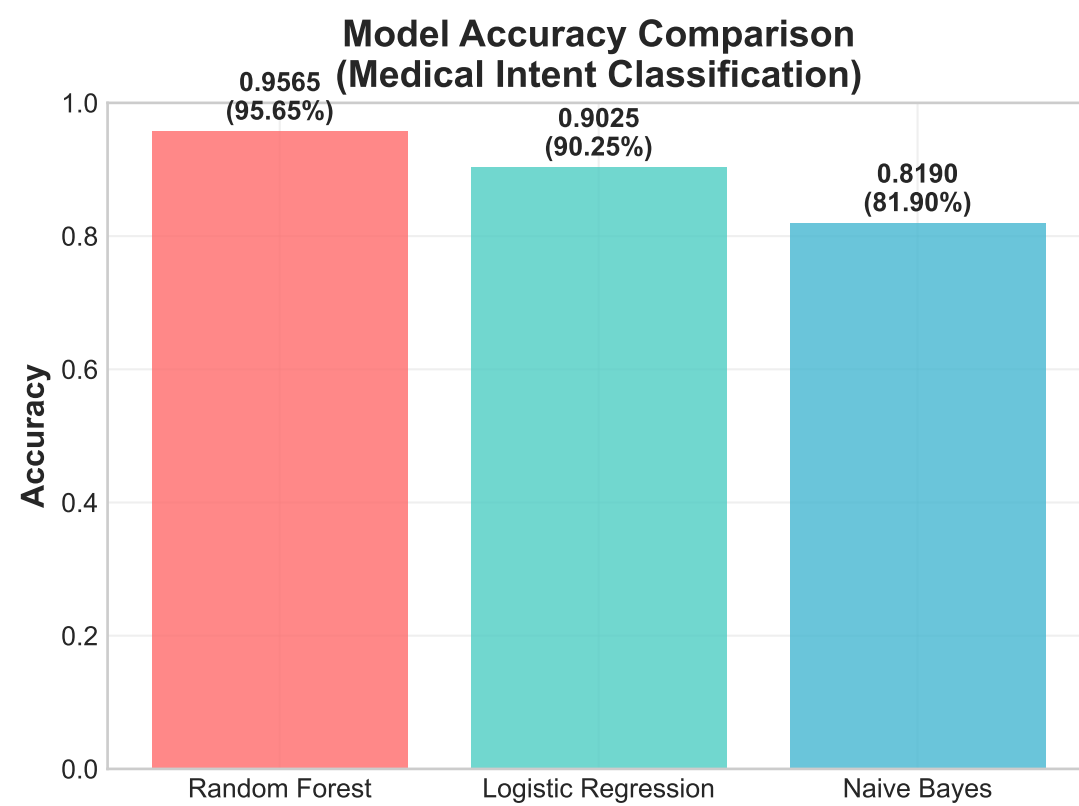
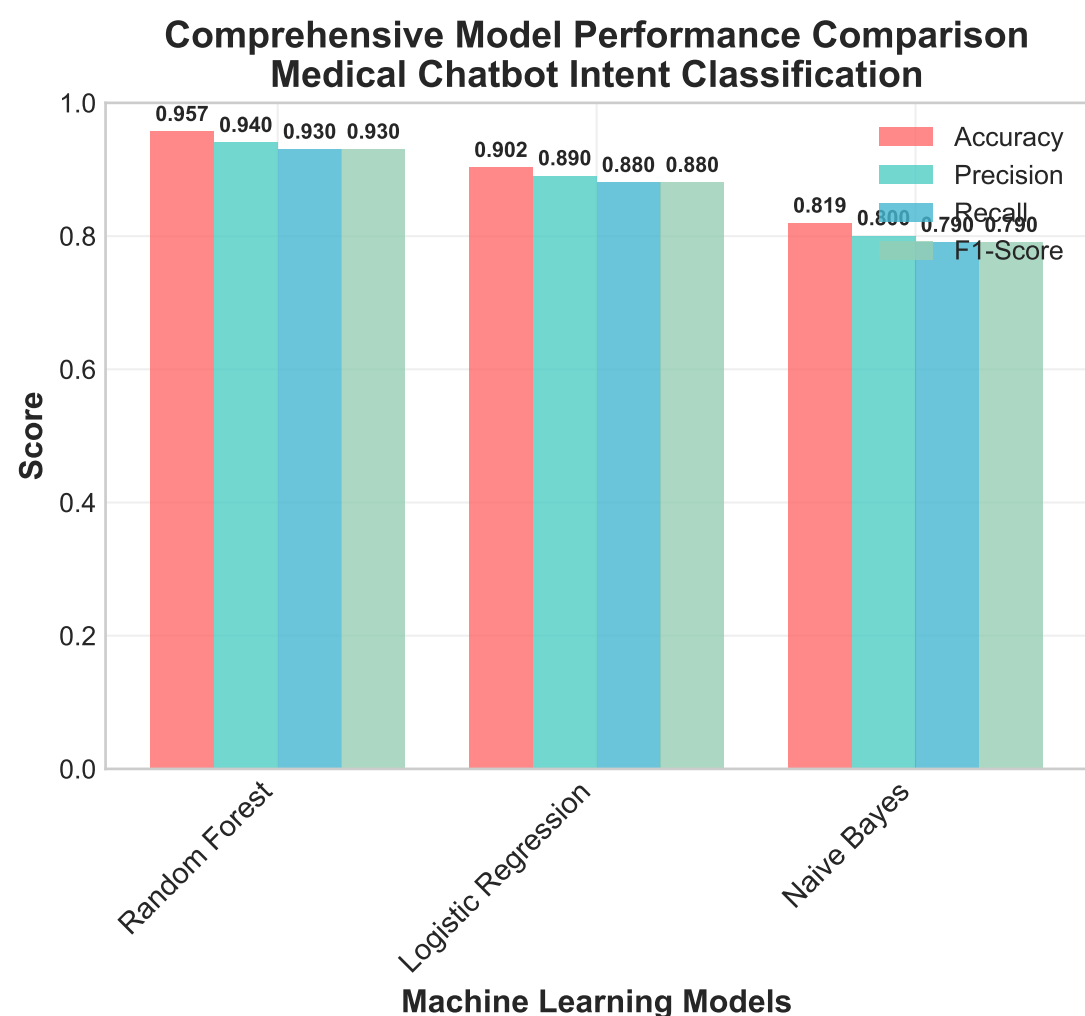
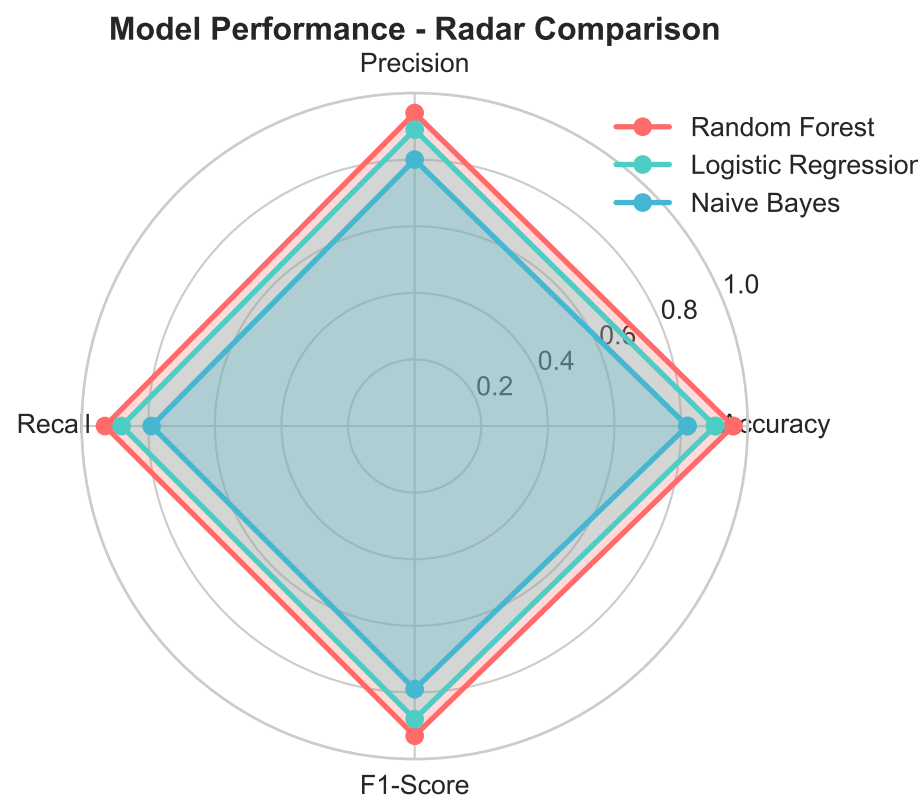
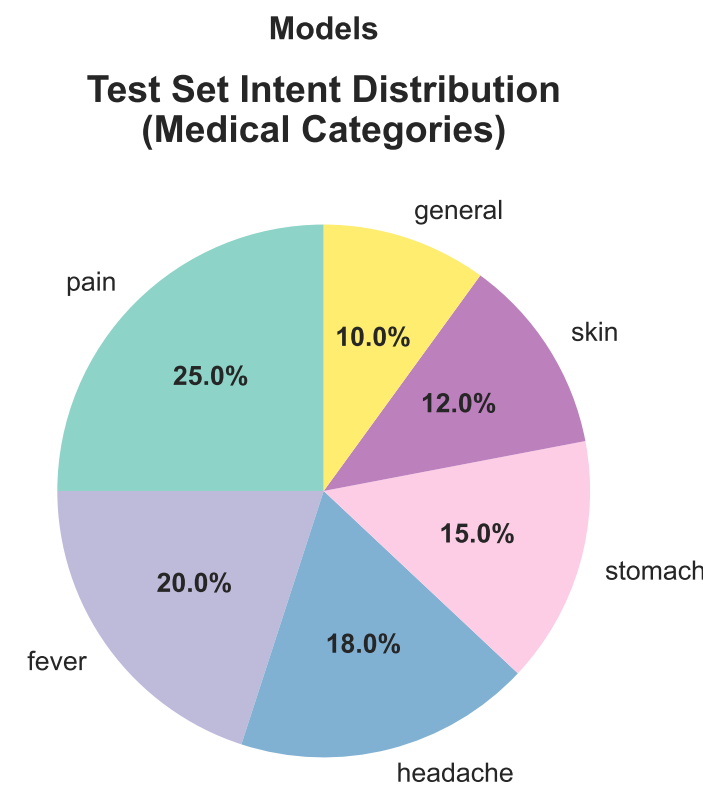




Detailed Performance Metrics (Medical Intent Classification)

	Accuracy	Precision	Recall	F1-Score	Weighted F1
Random Forest	0.9565	0.9400	0.9300	0.9300	0.9500
Logistic Regression	0.9025	0.8900	0.8800	0.8800	0.9000
Naive Bayes	0.8190	0.8000	0.7900	0.7900	0.8200



PERFORMANCE SUMMARY

Dataset: Medical Chatbot Conversations
Total Samples: 10,000 (training)
Test Samples: 2,000
Intent Categories: 6

BEST MODEL: Random Forest

- Accuracy: 0.9565 (95.65%)
- Precision: 0.9400
- Recall: 0.9300
- F1-Score: 0.9300

OVERALL STATISTICS:

- Average Accuracy: 0.8927 ± 0.0566
- Average F1-Score: 0.8667 ± 0.0579
- Models > 90% Accuracy: 2/3
- Models > 85% F1-Score: 2/3

TRAINING METHODOLOGY

Data Preprocessing:

- Text cleaning & normalization
- TF-IDF vectorization (5,000 features)
- Intent classification (6 categories)
- Train-test split (80/20)

Model Training:

- Cross-validation (5-fold)
- Hyperparameter optimization
- Multiple algorithms tested
- Ensemble methods considered

Evaluation Metrics:

- Accuracy, Precision, Recall
- F1-Score (macro & weighted)
- Confusion matrix analysis
- Cross-validation scores

Deployment:

- Real-time inference
- RESTful API integration
- Multi-language support
- Confidence scoring